FMI ADMINISTRATIVE INFORMATION	FMI No: 25394

FMI Title: 07MW18.4 VCT Quality Release

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☑ Mandatory Safety	GEMS – AM:	Issue Date:
☐ Mandatory	GEMS – E:	
☐ Mandatory on Request	GEMS – A:	Close Date:

☑ **Staged FMI** — **Customer List** provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole.

☑ **Parts** and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole.

☐ **Parts Ordering**: GEMS - Americas, Contact your Region logistics coordinator
GEMS - Europe, Order FMI kits from GPO/EPL Buc
GEMS - Asia, Contact your region logistics coordinator

☐ **No Parts** are required to complete this FMI.

☐ **Batch with** ______________________________

To minimize costs, combine implementation of this FMI with the FMI's listed for all customers where the effectivity is the same (Please see your customer lists).

☐ **Non-Staged** — **No Customer List is provided**. Each service location must identify affected units, and place part orders as indicated in the Special Instructions.

Technical Support

GEMS – Americas	GEMS - Europe	GEMS – Asia
GE Medical Systems **On-Line Center** **800-321-7937** **Milwaukee, WI 53201, USA**	**GEMS - E / ESC** **BP34** **F 78533 BUC CEDEX, FRANCE** **Tel: (33 1) 39 20 00 07 or ESC** **Fax: (33 1) 30 70 99 70**	**Asia Support Center** **Phone: 81-426-56-0033** **Fax: 81-426-48-2905**

Special Instructions:

This FMI does not apply to CT/PET Systems! If you received this kit for a CT/PET system, STOP. Close FMI as Code 2 and return kit.

This FMI includes an LFC procedure which will be accessed from service CD 5193574-200 included in the kit. See Procedure in Section 4 of this document for details!

If you have an English Speaking site (selected English Keyboard only, GEHC2 during LFC), Service Pack 1.5 CD is not required.

Unit Tracked: **Multiple VCT Gantries**

FMI Instructions Part No.: **5195239-100 Rev 5**

Do NOT leave unsecured on site

FMI Completion Certification Report

Mail To

GEMS - Americas	GEMS – Europe	
G.E. Medical Systems Attn: FMI Admin. W-523 P.O. Box 414 Milwaukee, WI 53201	GEMS - Europe FMI Admin. 322 BP34 F78533 Buc Cedex, France	Send to your Country FMI or Service Administrator

FMI No.: 25394 FMI Title: O7MW18.4 VCT Quality Release

Customer Full Name:

Number and Street Address:

City and Country:

Country Code Site Number System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code

1. Modification Complete or previously modified.
2. FMI not required per FMI instruction.
3. Not modified. Unit in storage. FMI kit returned to stock.
4. Modification refused or equipment altered. Customer signature and title required.
5. Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer Comments

Revision History

Revision	Date	Reason for Change
1	5/18/2007	Initial Release
2	6/1/2007	Added section 5.3; other minor corrections.
3	6/22/2007	Update furnished material (page 3) and other minor corrections based on F3 feedback
4	7/24/2007	Updates to software load instructions and BIOS update
5	8/28/2007	Added of Servicepack 1.5 (to fix voxtool problem) and load instructions

Effectivity

The FMI effectivity includes CT VCT64, VCT32 and VCTSTD systems. The following are tracked gantry model numbers. This does not apply to CTPET system gantries.

Gantry Model Numbers:

- 5124069
- 5124069-2
- 5124069-5
- 5124069-6
- 5126093

Purpose

The purpose of this FMI is to upgrade all VCT systems with the new software 07MW18.4.

This software release improves the quality of all VCT systems and also introduces several new features. A brief description of the key fixes and features are given in ‘Product Fixes’ and ‘New Features’ sections.

Prerequisites

1 Software:

Please make sure that the software revisions are latest for your system (as shown below) before loading the new software. If not, update the system before proceeding with this FMI.

System	Software Version
VCT64	06MW03.4* or 06MW29.7
VCT32	06MW03.4* or 06MW29.7
VCTSTD	06MW03.4* or 06MW29.7

*The 06MW03.4 was delivered through FMI 25373 to the installed base.

To check software version:

As a ctuser, type **swhwinfo** <Enter>

2 Table Belt Tension Adjustment:

FMI 25400 should be done prior to or along with FMI 25394.

3 BIOS update:

HP BIOS Settings. This will be done during the LFC procedure, no need to install at this step.

HP8200 - This should be at BIOS 2.10 - A BIOS 2.10 Floppy Disk and Configuration Floppy have been included in the kit. This procedure is addressed in the LFC procedure.

HP8000 - This should be at BIOS 1.13 - No change in BIOS will be required.

Note: An additional BIOS Disk can be created if needed by downloading from Support Central at the following link. BIOS disk can be created over the existing disk or on a new one. It is advisable that FE's load the BIOS and read-me file onto their laptop prior to loading software. *CT Online Support / Tools / Patch Downloads / HP Computer Bios*

http://supportcentral.ge.com/products/sup_products.asp?prod_id=15768

Furnished Materials

Below are the items included in kit# 5195244 Rev 2.

Item #	Qty	Description	Part #	Rev
1	1	FMI Instructions	5195239-100	5
2	1	BIAS Service Tool Assembly	5134603	3
3	1	DVD RAM Media, 9.4 GB (Blank Disk)	2337414	2
4**	1	BIOS 2.10 Floppy Set, BIOS Floppy and Setup Floppy - Floppy Disk, xw8200 BIOS 2.10, 5191796, Rev 1 - Floppy Disk, xw8200 BIOS 2.10 Configuration Disk, 5191796-2, Rev 1	5191796-3	1
5**	1	Software Collector and with Rating Plate - Software Collector, 5212369, Rev 1 o Applications Software, 5212589 Rev 2 o Operating System Software, 5212704, Rev 1 o Serial Over LAN software, 5212235, Rev 1 - Rating Plate, 46-302200P7, Rev 4	5213049	1
6**	1	Service Pack 1.5 CDROM with product locator	5234959-15	2
7**	1	LightSpeed 7.X Service Information and LFC	5193574-200	4
8**	1	LightSpeed VCT Learning and Reference Guide	5119850-247	7
9**	1	LightSpeed VCT Quick Steps Guide	5134008-100	7
10**	1	LightSpeed VCT Quick Steps Guide	5134008-247	7
11**	1	LightSpeed 7.X Applications T & W	5116420-100	9
12**	1	LightSpeed 7.X Applications T & W	5116420-247	9
13**	1	LightSpeed VCT Cardiac Scanning Guidelines for Coronary Artery Imaging	5166167-100	3
14**	1	LightSpeed VCT Cardiac Scanning Guidelines for Coronary Artery Imaging	5166167-247	3
15**	1	LightSpeed 7.X Technical Reference Manual - Multilanguage	5116422-247	9
16**	1	LightSpeed 7.X Technical Reference Manual - English	5116422-100	9
17**	1	Advanced Applications on the OC training disk	5183783-247	1
18**	1	LightSpeed 7.X Applications T & W - Chinese	5116420-141	9
19**	1	LightSpeed 7.X Technical Reference Manual – Chinese	5116422-141	9
20**	1	LightSpeed VCT Quick Steps Guide – Chinese	5134008-141	7
21**	1	Product Matrix Manual	5231794-148	1
22**	1	LightSpeed VCT Quick Steps Guide - Japanese	5134008-140	7

** All these items are in 5213243, rev 5, 'VCT Cardiac SW and Doc Collector'.

Tools Required

- LightSpeed Class C Software (tab 182 distribution) Part # 5220042
- LightSpeed Class M Software (tab 182 distribution) Part # 5213266
  (Class C and Class M software were distributed via tab distribution)
- Standard Service Tool Kit
- Customer purchased options

Personnel Requirements

Travel: 1 Hour, Labor: 7 hours Field Engineer: 1

Product Fixes

There are over 200 PSR's and PQR's fixed with this software release, dramatically improving quality. The most prominent fixes are described below.

X I/O Error:

Fatal error that occurs as a result of X* Server shuttting down a X client's connection. Processes such as scanRx, msgView, examRxDisplayMgr which are also X clients have been reported in the past to have crashed due to this error resulting in system shutdown. This error has been linked to bugs in both xorg implementation and NVIDIA graphic drivers. This has been fixed.

*X library allows programming GUIs on linux and other unix like systems.

Pinwheel Artifact Reduction:

To reduce the pinwheel artifacts in helical scans for slice thicknesses of 0.625 and 1.25mm, for all pitches, all kernels, all helical scan speeds, full and plus modes (z-smoothing in the image space).

ECG Viewer/R-Peak Editor - Option:

Provides edit capabilities on the ECG waveform display for Cardiac Helical and Cine modes in Retro Recon. User can insert, delete or move a trigger point and invert the trace, make measurements, magnify/minify trace, undo actions and save changes.

Daylight Savings Time Change:

The recent change in daylight saving time (now to begin second Sunday in March and end the first Sunday in November) for affected states have been implemented in this software release for VCT systems. There is no need for software patch, 5212541.

Bright shoulder Artifact Reduction from 3D Scatter Correction:

Improvements made in 3D Scatter Correction for elimination of streak artifact at the shoulder area on small patients.

Software Load Performance Improvements:

- Save/Restore: Optimized to cut down time up to 10 minutes
- Time Drift: Auto Time Adjust functionality to avoid time drift. Eliminates need for future setdates
- SOL: Improved to handle Alternate IP addresses correctly
- BIS: Enable/disable status saved as part of System State and restored appropriately.
- Flash Download: Elimination of one Flash Download by restructuring LFC procedure.

BIS Correction:

BIS (Bone Induced Spectra) is an image space correction for Bone induced artifacts. This only works for head bowtie when enabled. These artifacts are typically produced toward the center of an image when scanning heads or anatomy that has highly attenuation material.

New Features:

Service Features:

Image Artifact IQ Snap:

This function is designed to capture core system/scan/recon configuration and usage information at the time an Image Quality problem is found at a customer system. The data captured at the time of the artifact can be loaded onto a DVD and shipped to HQ for analysis.

Cardiac RTS:

This function is designed to gather run time statistics (RTS) for Cardiac exams. This data collected during applications scanning, or aborts, does not affect system performance.

Gantry Accuracy Improvement:

The gantry accuracy has been improved from 0.25 degrees to 0.14 degrees at zero degree tilt position.

This helps to reduce the Stair Stepping artifact.

Product Features Overview:

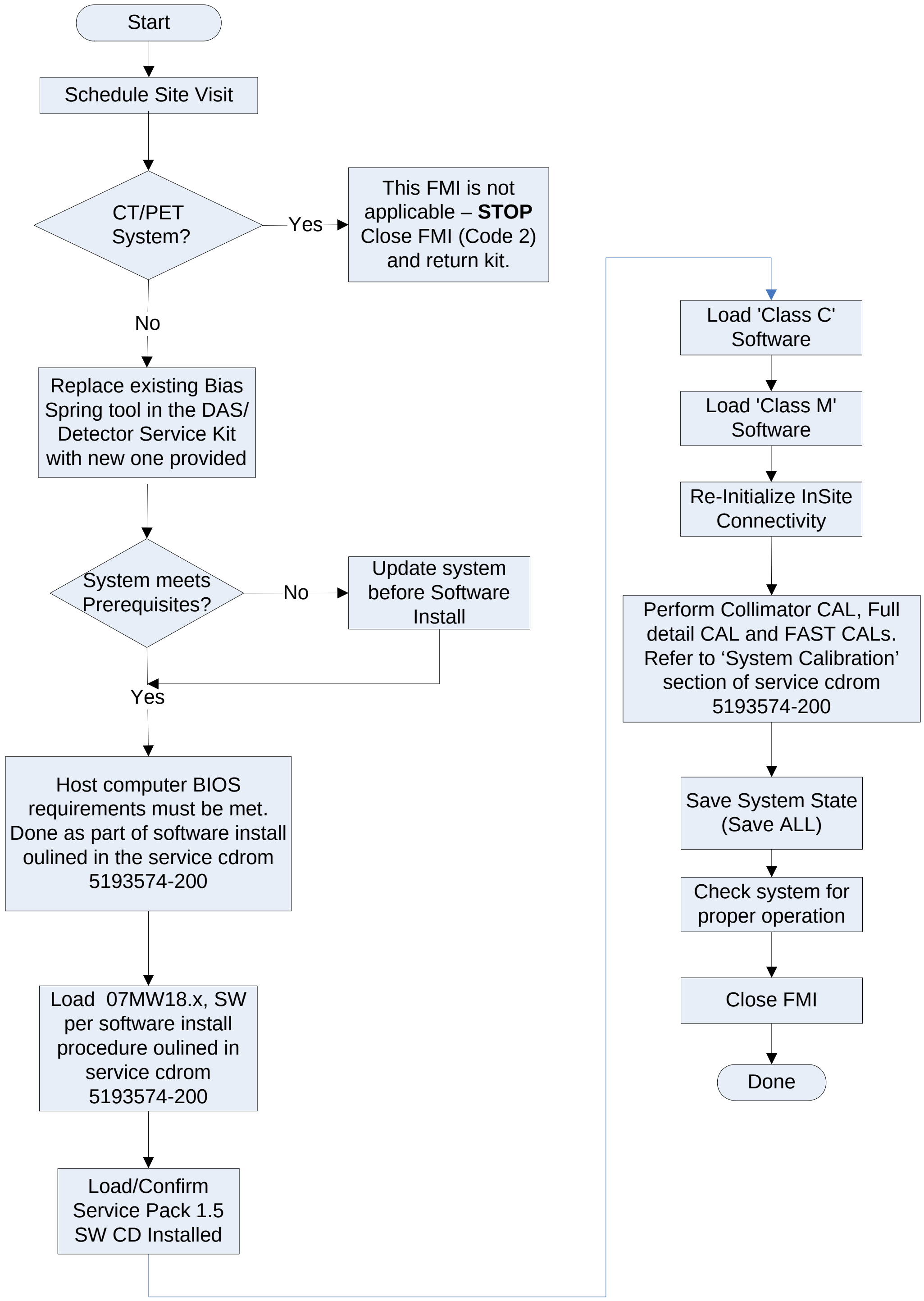
More details can be found in LightSpeed VCT Learning and Reference Manual – Direction 5119850 Rev 7.

- **Tube mA:** Tube mA will be limited if Tube-Warm-Up is skipped or cancelled. The mA will be limited to 250mA for small focal spot and 500mA for large focal spot in exam following a skipped or cancelled Tube Warm-Up. Although mA is limited, user can still achieve adequate mAs by changing rotation times or doing a Tube-Warm-Up. This protects the tube from selecting high mA when the tube is cold.
- **AutomA/SmartmA:** Changes have been implemented to provide more consistent behavior across varied patient sizes. The maximum Noise Index limit has been increased to 70.
- **Patient Schedule Update:** Revised to provide easier update of the patient schedule list.
- **Auto Voice:** Preset delays now support 0-7 second delays of 'x-ray on' after the end of the auto voice message.
- **SmartPrep:** Changes have been made to improve reliability. For multi-IG systems, all IGs locked for SmartPrep. AutoView for any reconstruction in the recon queue during SmartPrep series will be cancelled. Only images from the SmarPrep series will be seen in AutoView and series created after the SmarPrep series.

  AutoScan mode is automatically enabled whenever SmarPrep is selected for an acquisition. Message displayed in console information bar indicating AutoScan is enabled if it was off.
- **Contrast Presets:** IV and GI (Oral) contrast information can now be entered directly or selected from preset files for IV and Oral contrast. Presets are saved as part of Save State.
- **Auto Transfer by Series:** Enhanced to include Scouts, Dose Report, PMR auto Transfer for Recon 1, 2 or 3. You can select unto 4 hosts.
- **Prospective Multi-Recon – PMR:** PMR supports prescription for Recon 2 and Recon 3 for Cardiac Helical modes (Snapshot Segment, Snapshot Burst, Snapshot Segment Plus).
- **Iterative Bone Option (IBO) and Axial IQ Improvements:** Changes have been made for better visualization of structures at the bone/brain interface and the Axial and Cine Full IQ. Retro Recon supports recon of 0.625 and 1.25 for VCT/VCT XT and 1.25 mm for Pro^{32}/VCT Select for 40 mm detector coverage.
- **Large SFOV 3D Scatter Correction Improvements:** Improvements in 3D Scatter correction for Large DFOV to eliminate streak artifact at the shoulder area on small patients.
- **Tool Bar:**
  Unix Shell: Unix shell now can be displayed on either the right or left monitor.
  IQ Snap: Provides capability to reserve scan data for images with IQ issues. Must first make images anonymous and select the Anonymous Exam. Storage for 10 snaps, oldest one deleted if exceeded.
  Anonymous Patient Level: Toggles between Full and Partial anonymous levels.
- **Bone Plus and Edge Reconstruction Algorithms:** Modified to reduce aliasing artifacts seen when using these reconstruction algorithms (kernels) and 0.5:1 pitch.
- **Patient Orientation – AutomA:** If patient orientation is different from the protocol, AutomA will be disabled. Once the orientation matches, AutomA can be enabled again.
- **Direct Vis – Auto Apps:** Direct Vis has been renamed Auto Apps to support the Neuro 3D filter and Auto Filter option for Neuro 3D.

- **Priority Recon:** Priority Recon will be disabled for some acquisition modes such as Snapshot Pulse or series where IQ Enhance has been selected. The Priority Recon button will be insensitive or sensitive depending on the parameters selected.
- **Reformat Changes:** This change allows the user define the series description for reformat series. Name Batch Series is enabled under Fill/Save Options.
- **Large Annotation Font:** Includes date for display on screen or film. Saved as part of System State.
- **Save on Screen Changes:** This allows the user to define the series description for 3D images.
- **Image Works – Screen Save Exam/Series Text Pages:** Exam and Series Text pages can be screen saved to be archived or sent to PACS.
- **Image Enhance:** This is a special reconstruction mode for helical scans that can only be selected for 0.625 mm and 1.25 mm slice thicknesses to minimize image artifacts commonly seen in thin slice helical acquisitions. Time to first image and image-to-image times will increase slightly for acquisitions when Image Enhance is enabled.

Process Flow



Procedure

1. Preliminary Checks

If this is CT/PET system, this FMI is not applicable. STOP. Close FMI as Code 2 and return kit.

2. DAS Detector Bias tool (BIAS tool assembly)

Existing Bias Spring tool in the DAS/Detector service kit may not have the magnets attached. Replace the Bias Spring tool in the DAS/Detector service kit with the BIAS Tool, 5134603, supplied with this FMI.

3. Before the upgrade, record the following information:

- **InSite™ Connectivity Information:**
  - Open Unix shell
  - Type **cd /usr/g/insite** <Enter>
  - Type **more .insiteINFO** <Enter>

Record connectivity configuration parameters.

- **VCT Detector Center Module (BIS, Bone Induced Spectra) Feature**

Save and Restore BIS Status:

For systems with 06MW03.4 software:

No action needed. Proceed to Step 3.

For systems with 06MW29.7 software:

BIS, center test and calibration vectors feature, is introduced with 06MW29.7. This feature allows image space correction for Bone induced artifacts. This feature is enabled only after a center module has been replaced (Service Note SN05091210d). Once the BIS status is enabled, it stays enabled. 06MW29.7 software **does not** save BIS status in System State files. The following procedure outlines how to find the status and then restore it after the new software load. There is no need to do the checking and restoring in the future.

Finding the status:

- Open Unix shell
- As a ctuser, type **enableBISfeature** <Enter>

If BIS is enabled, you will see the following output:

info - BIS is already enabled in the image reconstruction subsystem

****
** *the applications MUST be restarted to finish enabling BIS* **
****

In this case, record that BIS is enabled. **This command must be executed again after the Load From Cold procedure to enable BIS status (section 9).**

If BIS is not enabled, you will see the following message:

info - enabling BIS in the image reconstruction subsystem

****
** *the applications MUST be restarted to finish enabling BIS* **
****

There is no action needed if BIS is not enabled.

4. Software Load Procedure

4.1 Need-To-Know before performing LFC

1. If your system has CT Colono Option (B7868LC), when you tried to reload options after loading software, the CT Colono Option will fail to load. The new software does not support CT Colono Option (B7868LC). If that happens, please contact your OLC to receive information on **free upgrade** to Advantage CT Colonography Plus (B7864KM). **When you call the support center, please provide both System ID and the system order number so that support center can check and issue eLicense fast.**

   Options loaded via e-License will be saved to State on future saves, therefore, it is recommended to complete a Save State operation shortly after loading this option.

2. **Section 7.2 Operating System Software Installation on Host Computer for *<u>VDARC</u>***

   The last paragraph of the section reads as follows:

   If the DARC BIOS is changed, it must be returned to original settings after the VDARC Node software load is complete for proper normal operation. Failure to return the VDARC Node Boot Order BIOS after the load process is complete may result in VDARC boot issues.

   <u>NOTE:</u>

   This paragraph is incorrect. If DARC BIOS on VCT is changed, it <u>DOES NOT</u> have to be returned to its original setting.

3. Section 8: Service Software

The step indicates the following:

Service software applies only to GE Healthcare personnel and customers with an Advance Service Limited License agreement. Load service software per the instructions on the media, if applicable.

This statement means that Class C and Class M software should be loaded at this time. Class C and Class M instructions are located in step 5.1 and 5.2 of this FMI instruction and if completed during the LFC process, does not need to be reloaded after returning to the FMI document after the LFC.

4. IVY Cardiac Trigger Monitor

If a system has **purchased** the Snapshot Pulse Option at the time that 07MW18.4 software is loaded, it must have the Ivy Cardiac Trigger Monitor model # 3150-A. Both the 3150 and the 3150-A models will have model 3150 on the front of the monitor. Model 3150-A monitors should have label on the back in the upper left-hand corner. Not all of 3150-A monitors will have the label on the back, therefore, a secondary check should be made of the sides of the monitor. On the 3150-A, one side will have a Floppy Disk drive, the other side will have a build-in Chart Recorder.

For Customers that purchase Snapshot Pulse in the future, the Cardiac Trigger Monitor will be addressed at the time of the order, so there is no need to upgrade at this time.

5. Autovoice Volume:

Prior to this software release, the Autovoice was not working as expected and there is a strong possibility that the Autovoice settings were changed erratically / randomly to get the desired effect. With this software release, the Autovoice behavior was corrected to ensure the settings are read correctly from the audio settings file. If you experience Autovoice issues, see Appendix C for further information.

6. VoxTool / Reformat screens to be properly translated:

An issue exists with the use of the Reformat and Volume Viewer Applications (VoxTool) for some languages other than English. This problem will create a mix of English and the selected language to be displayed on the screen when using the Volume Viewer (VoxTool) / Reformat Screens.

To correct this issue, Service Pack 1.5 CD (p/n 5234957-15 inside software collector p/n 5234959-15) has been developed and deployed with the latest revision of FMI 25394 kit. Service Pack 1.5 should be loaded after completing the LFC procedure. Please refer to section 5 of this document for instructions on loading Service Pack 1.5.

NOTE:

If you have an English Speaking site (selected English Keyboard only, GEHC2 during LFC), this Service Pack 1.5 CD is not required.

7. **Prospective Multi-Recon – PMR:** This feature is not currently functioning as designed. This will be corrected with a future Service Pack Release. Please tell customer not to use Prospective Multi-Recon feature at this time.

   As a workaround, only use Retro Recon for reconstruction of additional phase for cardiac acquisitions.

4.2 Performing LFC

Perform software load using the procedure outlined in serviced cdrom 5193574-200 Rev 4 or later.

1. Open CDROM 5193574-200 ‘LS 7.X Limited Access Service Information (Gen)
2. Click on Software folder.
3. Click on Software Installation Procedures (LFC) folder.
4. Click on 07MW18.x SW LFC (LS7.X) procedure and follow the instructions.

5. Service Pack 1.5 Software Load

Service Pack 1.5 CDROM (p/n 5234957-15 inside software collector p/n 5234959-15 shipped in this FMI kit) must be installed for VoxTool / Reformat screens to be properly translated into the language for which the system has been configured.

a) Retrieve Service Pack 1.5 CDROM, p/n 5234957-15 inside software collector p/n 5234959-15, from the FMI kit

b) Insert the CDROM into the HP PC DVD drive

c) Shutdown Apps

d) Open Unix shell

e) Login: **su –** <Enter>

f) Type: **patch_install –c** <Enter>

g) Type: **reboot** <Enter>

h) Type: **showprods | grep -i servicepack** <Enter>

   Example of returned string:

   07MW11.10_07MW18.4_07BW08.4_GEHCServicePack1.5 08/20/2007
   07MW11.10/07MW18.4/07BW08.4 Service Pack 1.5 ver 1 rel 1.2

6. Load Class C and Class M Software

To ensure proper load of the InSite™ software, please follow the following sequence:

1. Load C Software
2. Load M software
3. Re-initialize InSite™ Connectivity

If the Class C and Class M Software install log exhibits errors, usually a repeat attempt at loading the software succeeds.

If Class C and Class M software have already been loaded as part of the Load From Cold procedure, proceed to Section 6 Re-Initializing InSite™ Connectivity.

6.1 Load Class C Software

Load Class C software as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su –** <Enter>
   b. At the Password prompt, type: **#bigguy** <Enter>
3. Insert the Class C software CD into HP computer DVD Drive. Wait for the disk read to STOP.
4. Install the Class C software on the system. At the root prompt, type: **install.classC** <Enter>
5. Close windows when message "Installation of Class C complete" appears.
6. Verify Class C is properly loaded on the system.
   a. Open a Unix Shell
   b. At the root prompt, type: **cd /var/adm** <Enter>
   c. At the root prompt, type: **ls** <Enter>
   d. Find the install.classc.log file. If file found, this means the Class C software load completed.
7. Remove the Class C software CD from HP computer DVD Drive.

6.2 Load Class M Software

If your system does not have InSite™ Entitlement, go to section 7 (on page 17).
Load provided Class M software as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su –** <Enter>
   b. At the Password prompt, type: **#bigguy** <Enter>
3. Insert the Class M software CD into the HP computer DVD drive. Wait for the disk read to STOP.
4. Install the Class M software on the system. At the root prompt, type: **install.classM** <Enter>
5. Close windows when message "Installation of Class M complete" appears.
6. Verify Class M is properly loaded on the system.
   a. Open a Unix Shell
   b. At the root prompt, type: **cd /var/adm** <Enter>
   c. At the root prompt, type: **ls** <Enter>
   d. Find the install.classM.log file. If file found, this means the Class M software load completed.
7. Remove the Class M software CD from HP computer DVD Drive.

6.3 Software Download / InSite Update File Ownership

To enable usage of Update Management buttons in Service Desktop/Utilities, the ownership of three files listed below must be set at this time as ctuser.

1. Open unix shell and verify that you are ctuser (not as root).
2. Type **touch /tmp/cl** <Enter>
3. Type **touch /tmp/cl1** <Enter>
4. Type **touch /tmp/cl2** <Enter>
5. To verify that ownership for cl, cl1, cl2 is now ctuser!

   Type **cd /tmp** <Enter>

   Type **ls -al cl*** <Enter>

   Output should look similar to below, note the ownership is ctuser.

```
-rw-rw-rw-  1 ctuser users  9 May 31 11:38 cl
-rw-rw-rw-  1 ctuser users  2 May 31 11:38 cl1
-rw-rw-rw-  1 ctuser users 27 May 31 11:38 cl2
-rw-rw-rw-  1 insite root  70 Jun  1 07:00 clientdrv.trace
```

6. Close unix shell.

7. Re-Initializing InSite™ Connectivity

7.1 Restore System State To Retrieve Class M Configuration Parameters

Restore 'Reconfig' part of the Restore System State to recover all previous InSite™ and Class M configuration parameters.

7.2 Re-Enable the ProDiag Utility

Enable the InSite™ Interactive Platform Configuration utility as follows:

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su –** <Enter>
   b. At the Password prompt, type: **#bigguy** <Enter>
3. Launch the IIP Admin Configuration utility. At the root prompt, type: **iipadmin config** <Enter>

   The system displays the InSite™ Interactive Platform Configuration utility with the Proprietary Agreement Acceptance Screen.
4. Click on ACCEPT three times to display the Information Screen.

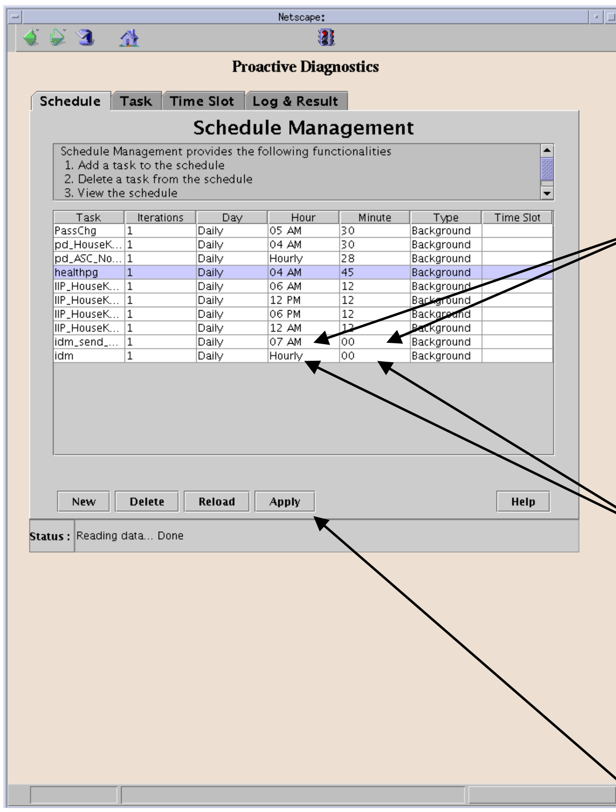
5. Click on the **ProDiags** Tab to display the ProDiags Screen.

   Click on the **Custom** button to launch the independent ProDiags

   This will launch ProDiag Utility**.**

6. There is no change to ProDiags. However, it is recommended that you review ProDiags Schedule and update, if needed, according to your particular site's normal operation. The Schedule Tab is used to modify Scheduled Tasks.

   As an example, use the following sequence to update the scheduled time (hour and minute) of the idm and idm_send_data tasks



1.

Click on **Hour** and **Minute** fields of idm_send_data task and select an hour and minute from the pull down lists between 9 AM and 5 PM.

After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

2.

Click on **Hour** and **Minute** fields of idm task and select an hour and minute from the pull down lists to schedule 5 minutes before the idm_send_data task.

After each selection, click **Yes** on the "Data has been modified. Please confirm it" pop up.

3.

When idm and idm_send_data Hour and Minute selections are complete, click on **Apply** to finalize the schedule upates.

Click **Yes** on the "Are you sure you want to apply changes?" pop up.

7. Click the minus sign at the top left browser window and a pull down window appears. Select **Close.** The ProDiags browser should close and the InSite™ Interactive Platform Configuration utility should display the HealthPage Tab as shown in Figure 5.

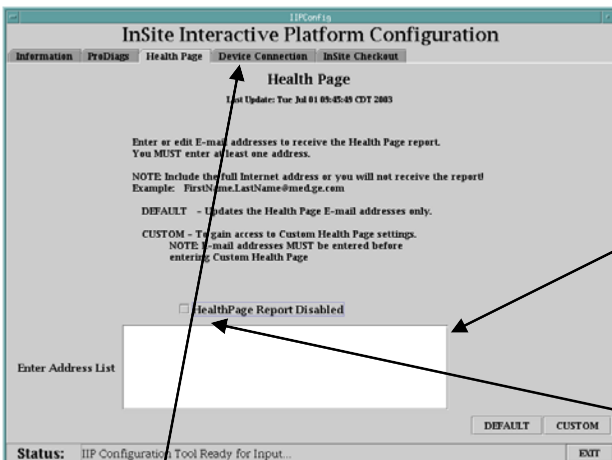
7.3 Configure HealthPage

After completing the ProDiags setup, the IIP Admin Configuration utility automatically switches to the HealthPage.

The Restore State process (after loading Class M software) restored the following information for the HealthPage Reports:

- All e-mail addresses of targeted recipients that were previously loaded
- HealthPage report contents that was previously configured
- HealthPage report schedule that was previously configured

Unless you have recent changes to any of these three items, you can skip the HealthPage Tab of the IIP Admin Configuration utility and select the Device Connection Tab. If you have changes that you want to make (e-mail addresses, report content, report schedule), follow the instructions on the IIP Admin Configuration utility.



HealthPage Screen

1. Enter all GE employee e-mail addresses using the following example:

   John.Doe@med.ge.com

2. If already not selected, click on the **HealthPage Report Disabled** box to enable HealthPage reports. Label changes to **HealthPage Report Enabled**.

3. Click on **Device Connection** Tab to check connection configuration information.

7.4 Check Device Connection Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen.

Make sure that the modem set-up parameters are correct. If not, change the parameters to the correct ones. See Appendix A.

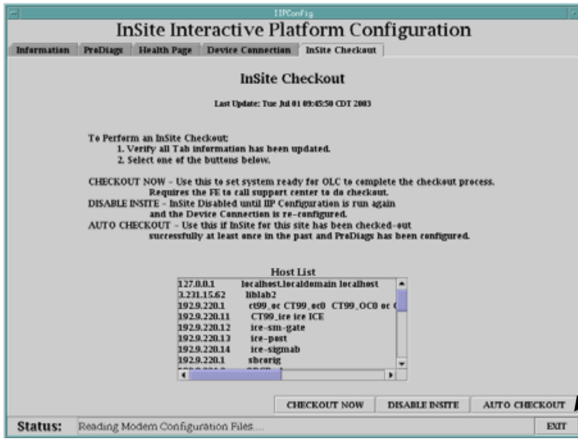
7.5 Check BroadBand Connection Setup

If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. If not, enter the proper IP address of the InSite™ Gateway. See Appendix B for instructions.

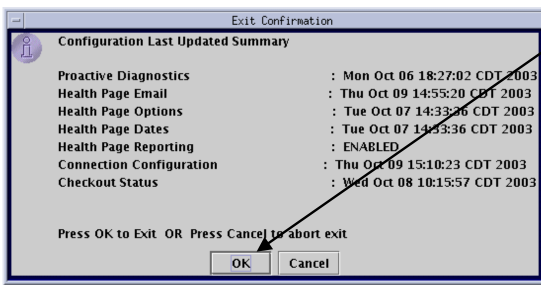
7.6 Completing the InSite™ Re-Initialization

Follow the instructions provided below to exit the IIP Admin Configuration Utility.

Insite Checkout Screen



1\. Click on the Exit button of the IIP Admin Config utility to exit the utility.



2\. Click on **OK** on the Exit Confirmation pop-up to exit the IIP Admin Config utility and run the ConfigLink utility.

3\. After the shell displays the root prompt, reboot the system to load Insite connectivity parameters. As the system is coming up, you should see the iLinq icon (if the original Checkout included iLinq) as a selectable desktop on the right console monitor.

7.7 Run ConfigLink

Run the InSite™ **ConfigLink** script to populate all configuration information required to fully restore connectivity parameters on the system.

1. Open a shell.
2. Enter root as superuser:
   a. At the ctuser prompt, type: **su –** <Enter>
   b. At the Password prompt, type: **#bigguy** <Enter>
3. Change directory to InSite™. At the root prompt, type:
   cd /usr/g/insite <Enter>
4. Run the ConfigLink script. At the root prompt, type:
   ConfigLink <Enter>

Wait for the shell to return the root prompt to ensure the **ConfigLink** process completes. When complete, exit root and close the shell.

8. Reboot System

1. Click on the pink Shutdown button.
2. From the Shutdown selection pop-up, click on RESTART, then OK, to reboot the system.

9. Calibrations

Due to the changes in recon with IQ improvements, the calibrations specified below are required. **It is mandatory at this time to complete the following calibrations:**

1. Collimator Cal
2. Detailed Cal
3. Fastcal

For detailed procedure, refer to 'System Calibration' section under 'Systems' section of 5193574-200 "LS 7.X Limited Access Service Information, Rev 4"

10. Restore BIS Status

Refer to section 3.

If BIS Status was not enabled, skip this section and go to section 10.

If BIS status was enabled, then execute the following to restore BIS Status:

Open a Unix shell

- As a ctuser, type **enableBISfeature** <Enter>

You will see the following output:

info - enabling BIS in the image reconstruction subsystem

****
** *the applications MUST be restarted to finish enabling BIS* **
****

11. Save System State

Save system state (**Save All**) using the Save System State utility.

12. System Checkout

Perform the following scans (Abdominal or Head)

50 images at 0.5 sec rotation at 120KV, 40mA

The system should scan and image without any errors

13. Finishing Up

Leave the following documents with the customer. Remove and discard all older revisions.

5119850-247	Rev 7	Learning and Reference Guide
5134008-100 and 5134008-247	Rev 7	Quick Step Guide
5116420-100 and 5116420-247	Rev 9	Tips and Workarounds
5166167-100 and 5166167-247	Rev 3	Cardiac Scanning Guidelines
5116422-100 and 5116422-247	Rev 9	Technical Reference Manual
5231794-148	Rev 1	Product Matrix

For systems in China:

Leave the following manuals in addition to the above:

5116420-141	Rev 9	Applications Tips and Workarounds - Chinese
5116422-141	Rev 9	Technical Reference Manual - Chinese
5134008-141	Rev 7	Quick Step Guide - Chinese

For systems in Japan:

Leave the following manual in addition to the above:

5134008-140	Rev 7	Quick Step Guide - Japanese

Discard any old revision documents.

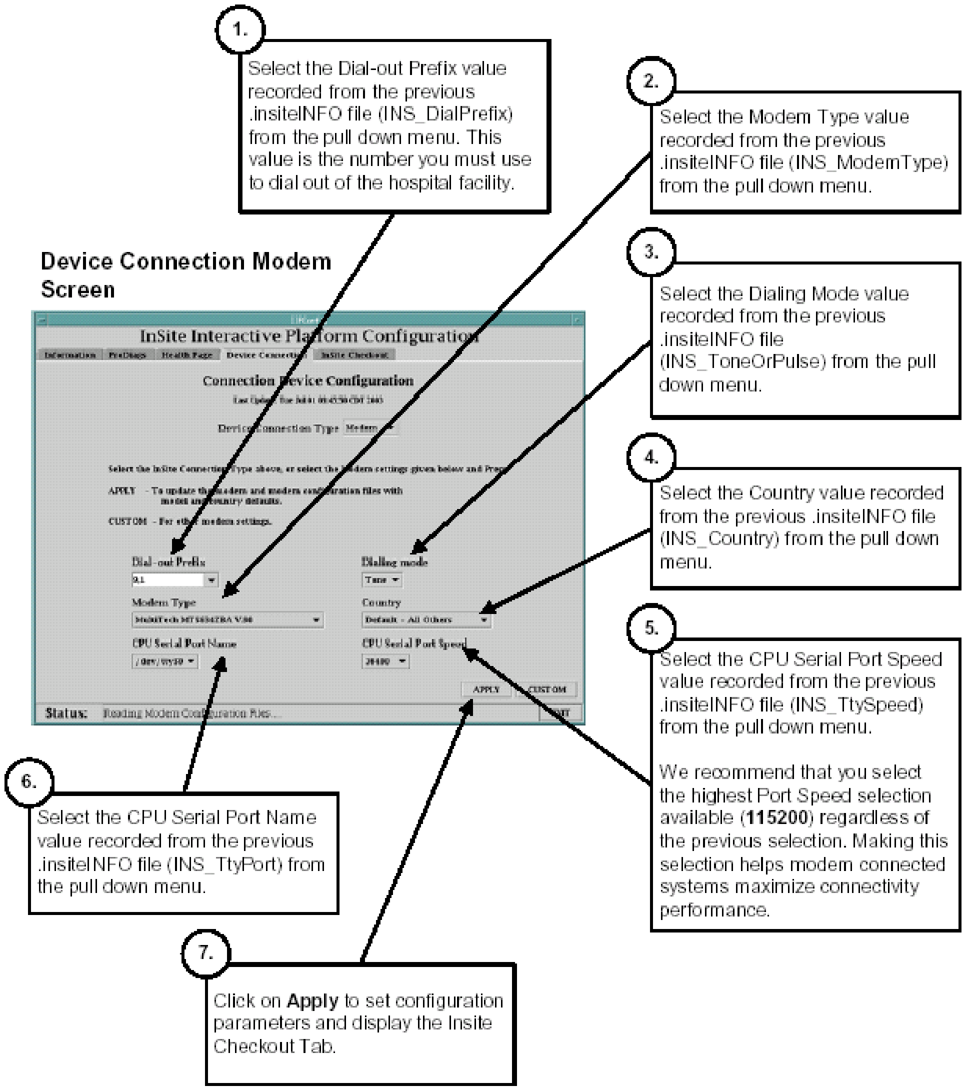
14. FMI Closure

Close FMI per regional procedure.

Remove FMI Instructions from the customer site.

APPENDIX A
Modem Setup

When selecting the Device Connection Tab, the IIP Admin Configuration utility automatically displays the Modem Setup Screen as shown below.



If your system was configured for Modem connectivity, the Restore State process (after loading Class M software) restored all the Modem setup parameters. Verify with the InSite™ information that you recorded before starting the loading process.

- If there were no Modem parameters in the .InSite™INFO file but there were BroadBand parameters, skip to the Check BroadBand Device Connection Setup procedure.
- If there were Modem parameters in the .InSite™INFO file but the parameters displayed in the Modem Setup Screen do not match what you previously recorded, enter the correct values as required.

NOTE: The Modem configuration parameters should be pre-loaded from the Restore State process. Do not change settings unless these values were not properly restored.

Appendix B
Broadband Connection Set Up

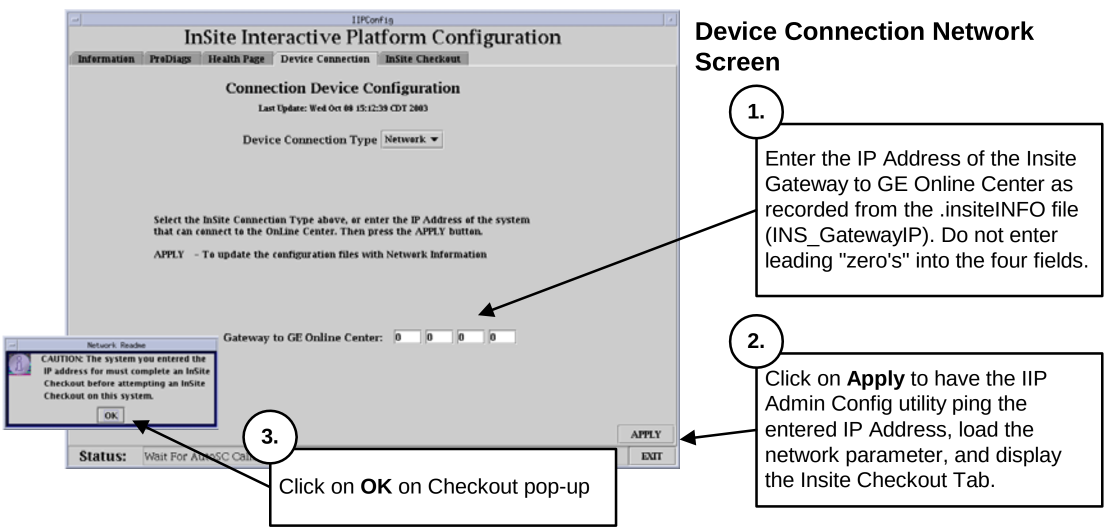
If your system was previously BroadBand InSite™ connected, click on the Network button of the IIP Admin Configuration utility Device Connection Tab to display the Network Setup Screen.

If your system was configured for BroadBand connectivity, the Restore State process (after loading Class M software) restored all the BroadBand setup parameters. If not, enter the proper IP address of the InSite™ Gateway as shown in the figure below.

NOTE: The BroadBand Network configuration parameters should be pre-loaded from the Restore State process. However, the on-site FE who performed the original InSite™ Checkout of the system may have used the hospital's imaging LAN default gateway IP Address (selection was available in the previous release of IIP).

DO NOT USE THE HOSPITAL WORKING LAN DEFAULT GATEWAY IP ADDRESS AS THE InSite™ GATEWAY IP ADDRESS OTHER THAN AS A TEMPORARY MEANS TO CONNECT TO THE ON LINE CENTER!! Contact the hospital network administrator to determine the right gateway address.

Device Connection Network Screen



Appendix C

Autovoice Set Up: Volume and Feedback Issues

It is possible that after loading this software, the system may exhibit two different problems related to Autovoice setting, Low Volume and / or Feedback (squealing or noise) through the speakers. If you are experiencing Low Volume, start at step 1. for Feedback issues start at step 2.

1. If Autovoice volume is too low, select "Autovoice Volume" tool ("audioSetting" tool) from the Tool Chest menu to increase it as needed, and click the "Save," then "OK" buttons to save to file.

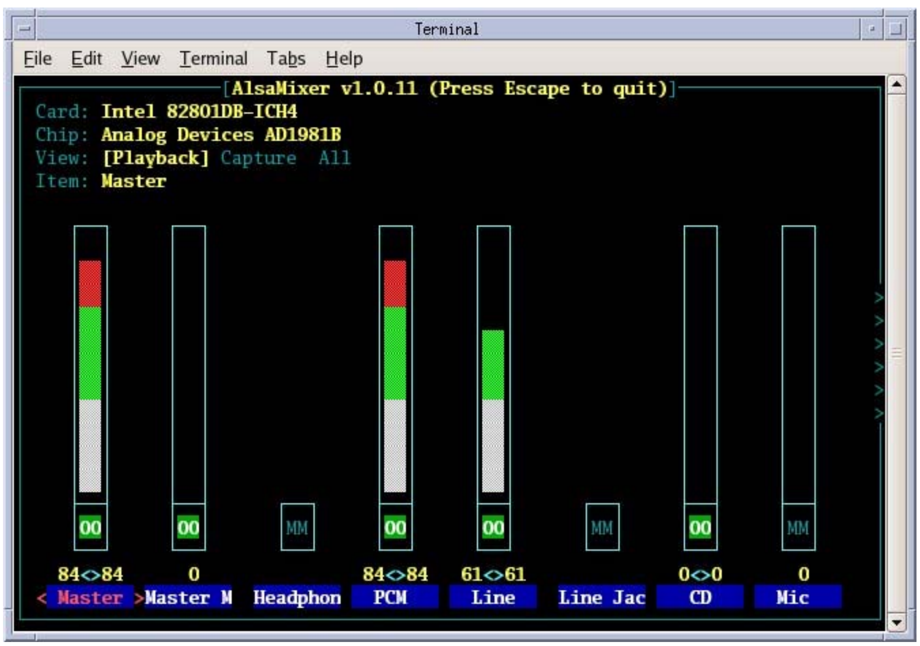
Changing Autovoice settings may induce feedback issues, if so, continue with step 2.

2. If Autovoice feedback issues exist, then

   a. Open a unix shell.

   b. Type: **alsamixer** [Enter]

   The following screen will appear:



NOTE: Settings shown here may not depict the final settings for your console. Settings vary from console to console.

   c. Press the right-arrow key → to move cursor selection to the **PCM** slider.

   d. Press the up-arrow key ↑ to increase the PCM setting to 100%.

e. If feedback is heard, press the right-arrow key → to move cursor selection to the **Line** slider.

   Press the down arrow key ↓ to decrease **Line** setting until feedback is no longer heard.

   Do not mute **Line** device; it is needed for Autovoice Record.

f. When completed, press the Esc key to exit AlsaMixer.

g. If the **Line** setting <u>was changed</u>, then STOP → you have completed Autovoice setup.

   If the **Line** setting <u>was not changed</u>, then select “Autovoice Volume” tool from Tool Chest menu and click the “Save,” then “OK” buttons to save the PCM settings to file.

NOTE: If **Line** setting was changed and then Autovoice Volume settings saved, the **Line** value set via AlsaMixer will be set back to “61;” it has been overwritten by this “Save.” Steps 2e and 2f must be repeated to adjust **Line** settings.